

Achyut Niroula

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EXPERIENCE

Nipissing University

Jan. 2026 – Apr. 2026

Software Developer, Honours Research Capstone

North Bay, ON

- Architected a cloud-native multimodal video understanding platform on AWS (EC2, S3, SQS, DynamoDB), integrating 8 neural models across distributed microservices.
- Built and owned REST API endpoints connecting async perception, fusion, and narrative services; fused vision, depth, motion, and audio into per-frame scene representations.
- Implemented post-deployment observability across all model endpoints; authored root-cause analysis reports and cross-service API documentation for engineering stakeholders.

Nipissing University

May 2023 – Aug. 2025

Web Developer, Research Assistant

North Bay, ON

- Built and maintained a responsive HTML5/CSS3/JavaScript website for a faculty member, integrated with a database-backed CMS and validated for accessibility each release.
- Gathered requirements from faculty stakeholders and delivered features across the full SDLC using Git branching and pull request workflows.
- Conducted peer code reviews and authored technical documentation for both developer and non-technical audiences.

The Home Depot

Dec. 2022 – Present

Sales Associate II, Flooring & Decor / Tool Rental

North Bay, ON

- Advised 20+ clients weekly on product and financing solutions across two specialized departments; resolved issues through clear verbal and written communication.
- Trained customers on tool operation and installation; adapted communication for audiences from contractors to first-time homeowners.

Seasonal Trades (AP Engineering, RussCollins Carpentry)

2023 – 2025

Construction Labourer / Design Consultant

North Bay, ON

- Performed site preparation and material handling on active construction sites; followed all safety procedures including PPE and lockout/tagout.
- Communicated design and layout concepts to clients that were frequently incorporated into project plans.

PROJECTS

LearnOne | *Java 21, Spring Boot, React, TypeScript, PostgreSQL, Redis, Claude API, pgvector*

- Built a Jarvis-style AI educator with persistent sessions, Claude API curriculum generation, SM-2 spaced repetition, Judge0 code execution, and KaTeX/Mermaid rendering.
- Designed a Spring Boot REST backend with PostgreSQL persistence, Redis caching, and a pgvector knowledge graph extracted from conversations.

AI Video Analyzer | *Python, FastAPI, React, AWS (EC2, S3, SQS, DynamoDB), Docker*

- Orchestrated 8+ neural models (SigLIP, DepthAnything V2, Mask2Former, SlowFast, ByteTrack, Whisper, Qwen2-VL) into a unified per-frame scene pipeline.
- Containerized all services on Linux/EC2; wrote 9 automated test suites covering frame-level correctness and edge cases.

ResOptimum | *Python, FastAPI, Flask, Claude API, pdfplumber, LaTeX*

- Built a resume optimizer that ingests a PDF and job description and generates an ATS-targeted LaTeX resume preserving factual accuracy.

- Engineered structured Claude API prompts with JSON schema validation, role detection, and a one-page LaTeX fit-check with auto-compression.

JobHunt AI | *Python, FastAPI, React, TypeScript, Docker, Nginx, SQLite*

- Built a multi-step LLM pipeline scraping 7 job boards concurrently with an ATS classifier and semantic resume matching end-to-end.
- Containerized deployment with Docker Compose and Nginx reverse proxy; wrote a pytest integration suite covering all API endpoints.

Cloud Inventory | *Python, Flask, Firebase, Google Cloud Run, Cloudinary, Chart.js, Docker*

- Deployed a serverless inventory system to Google Cloud Run with role-based access control (Admin/Editor/Viewer) and analytics dashboards.
- Integrated Firebase Realtime Database, Cloudinary CDN image uploads, and real-time data synchronization.

Cloud-Chat | *Java, Firebase Realtime Database, WebSocket, Maven*

- Built a distributed real-time messaging platform with WebSocket synchronization and event-driven listeners across distributed clients.
- Demonstrated CAP theorem and eventual consistency tradeoffs; wrote JUnit tests and Maven build configuration.

rlget | *Python, ThreadPoolExecutor, pytest, CLI*

- Built a production-style CLI file downloader with parallel downloads, token-bucket rate limiting, and exponential backoff with jitter.
- Handled real-world HTTP robustly with Content-Disposition parsing and no-clobber safe file naming; covered with pytest.

PromptShop Test Agent | *Python, OpenRouter API, OpenAI-compatible client*

- Built an LLM agent that reads structured JSON prompts and routes them to multiple open-source models via OpenRouter for automated evaluation.
- Designed for integration with agent execution systems using an OpenAI-compatible client.

Indeed Stealth Scraper | *Python, FastAPI, Playwright*

- Built a modular scraper with proxy rotation and exponential backoff exposing scrape and export endpoints via FastAPI.
- Wrote integration tests in live Playwright environments; documented root-cause analysis on bot-detection failures.

EDUCATION

Nipissing University

B.Sc. Honours Specialization in Computer Science | GPA: 3.7/4.0

May 2026

North Bay, ON

- President's Scholarship (2021); Carl Sanders Scholarship (2022–2025).
- Relevant coursework: Distributed Systems (96%), Databases & Data Management (96%), Software Engineering (89%), Security & Protection (86%), Data Structures (83%), Artificial Neural Networks (83%), Digital Systems (81%), Intro to Data Science (80%).

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, SQL, C++, HTML5/CSS3

Frameworks & APIs: FastAPI, Spring Boot, React, Next.js, Flask, Node.js, RESTful APIs, microservices, Nginx

AI & LLM: Claude API (Anthropic), OpenRouter, multi-step LLM pipelines, tool-calling, pgvector, RAG, semantic similarity, PyTorch models

Cloud & DevOps: AWS (EC2, S3, SQS, DynamoDB); Azure-equivalent architecture, Google Cloud Run, Firebase, Docker, Docker Compose, Git, CI/CD, Linux

Databases: PostgreSQL, Redis, SQLite, Firebase (NoSQL), pgvector

Testing: pytest, JUnit, Playwright, TDD, integration and end-to-end testing

Other: Stakeholder communication, technical writing, agile/SDLC, WHMIS certified